

Task 1

1. Created customers table in a notebook with same name

The first screenshot shows a Databricks notebook titled 'Customers' in Python. The notebook contains a code cell (Cmd 2) that reads a CSV file from the FileStore and creates a DataFrame. The code is as follows:

```
1 # File location and type
2 file_location = "/FileStore/tables/Customers-9.csv"
3 file_type = "csv"
4
5 # CSV options
6 infer_schema = "false"
7 first_row_is_header = "true"
8 delimiter = ","
9
10 # The applied options are for CSV files. For other file types, these will be ignored.
11 df = spark.read.format(file_type) \
12     .option("inferSchema", infer_schema) \
13     .option("header", first_row_is_header) \
14     .option("sep", delimiter) \
15     .load(file_location)
16
17 display(df)
```

The second screenshot shows the same notebook with a new code cell (Cmd 3) that creates a temporary view from the DataFrame. The code is as follows:

```
1 # Create a view or table
2
3 temp_table_name = "Customers_csv"
4
5 df.createOrReplaceTempView(temp_table_name)
```

Below the code cell, the command execution time is shown: "Command took 0.13 seconds -- by venkatagayathrip1498@gmail.com at 4/13/2023, 1:09:55 AM on gayathri".

The third screenshot shows a new code cell (Cmd 4) that queries the temporary view using SQL. The code is as follows:

```
1 %sql
2
3 /* Query the created temp table in a SQL cell */
4
5 select * from 'Customers_csv'
```

Below the code cell, the command execution time is shown: "Command took 0.52 seconds -- by venkatagayathrip1498@gmail.com at 4/13/2023, 1:09:59 AM on gayathri".

The output of the SQL query is displayed as a table:

customer_id	name	email	phone	location
1	John Doe	johndoe@example.com	555-555-5555	New York
2	Jane Doe	janedoe@example.com	555-555-5556	Los Angeles
3	Bob Smith	bobsmith@example.com	555-555-5557	Chicago

The table has 3 rows and 5 columns. The runtime is 0.52 seconds. The table was refreshed 9 minutes ago.

2. Created OrderItems table in notebook named 'Order Items'

community.cloud.databricks.com/?o=7575967165101867#notebook/2666332837658393/command/2666332837658398

Order Items Python

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```

1 %sql
2
3 /* Query the created temp table in a SQL cell */
4
5 select * from `OrderItems_csv`

```

(1) Spark Jobs

_sqldf: pyspark.sql.dataframe.DataFrame = [order_item_id: string, order_id: string ... 3 more fields]

Table +

	order_item_id	order_id	product_id	quantity	price
1	1	1001	1	2	20
2	2	1002	2	1	40
3	3	1003	1	3	20
4	4	1003	3	2	50

4 rows | 0.40 seconds runtime Refreshed 10 minutes ago

Command took 0.40 seconds -- by venkatagayathrip1498@gmail.com at 4/13/2023, 1:10:43 AM on gayathri1

```

Cmd 5
1 # With this registered as a temp view, it will only be available to this particular notebook. If you'd like other users to be able to query this
2 # table, you can also create a table from the DataFrame.
3 # Once saved, this table will persist across cluster restarts as well as allow various users across different notebooks to query this data.
4 # To do so, choose your table name and uncomment the bottom line.
5 permanent_table_name = "OrderItems_csv"
6
7 # df.write.format("parquet").saveAsTable(permanent_table_name)

```

3. Created Products table in this note book

community.cloud.databricks.com/?o=7575967165101867#notebook/2666332837658405/command/2666332837658410

Products Python

File Edit View Run Help Last edit was 9 minutes ago Give feedback

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Overview

This notebook will show you how to create and query a table or DataFrame that you uploaded to DBFS. DBFS is a Databricks File System that allows you to store data for querying inside of Databricks. This notebook assumes that you have a file already inside of DBFS that you would like to read from.

This notebook is written in **Python** so the default cell type is Python. However, you can use different languages by using the `%LANGUAGE` syntax. Python, Scala, SQL, and R are all supported.

```

Cmd 2
1 # File location and type
2 file_location = "/FileStore/tables/Products.csv"
3 file_type = "csv"
4
5 # CSV options
6 infer_schema = "false"
7 first_row_is_header = "true"
8 delimiter = ","
9
10 # The applied options are for CSV files. For other file types, these will be ignored.
11 df = spark.read.format(file_type) \
12     .option("inferSchema", infer_schema) \
13     .option("header", first_row_is_header) \
14     .option("sep", delimiter) \
15     .load(file_location)
16
17 display(df)

```

(2) Spark Jobs

df: pyspark.sql.dataframe.DataFrame = [product_id: string, product_name: string ... 2 more fields]

4. Created final table, orders with the notebook named as Orders

The screenshot shows a Databricks notebook interface. The browser address bar indicates the URL: `community.cloud.databricks.com/?o=7575967165101867#notebook/2666332837658411/command/2666332837658416`. The notebook is titled "Orders" and is in "Python" mode. The left sidebar shows the Databricks workspace with options like "Create", "Workspace", "Recents", "Search", "Data", "Compute", and "Workflows".

Overview

This notebook will show you how to create and query a table or DataFrame that you uploaded to DBFS. DBFS is a Databricks File System that allows you to store data for querying inside of Databricks. This notebook assumes that you have a file already inside of DBFS that you would like to read from.

This notebook is written in **Python** so the default cell type is Python. However, you can use different languages by using the `%LANGUAGE` syntax. Python, Scala, SQL, and R are all supported.

Code Cell (Cmd 2):

```
1 # File location and type
2 file_location = "/FileStore/tables/Orders-1.csv"
3 file_type = "csv"
4
5 # CSV options
6 infer_schema = "false"
7 first_row_is_header = "true"
8 delimiter = ","
9
10 # The applied options are for CSV files. For other file types, these will be ignored.
11 df = spark.read.format(file_type) \
12     .option("inferSchema", infer_schema) \
13     .option("header", first_row_is_header) \
14     .option("sep", delimiter) \
15     .load(file_location)
16
17 display(df)
```

Below the code, it shows "(2) Spark Jobs" and a preview of the DataFrame schema: `df: pyspark.sql.dataframe.DataFrame = [order_id: string, customer_id: string ... 2 more fields]`.

The bottom of the screen shows a Windows taskbar with various application icons, a search bar, and system status information including "62°F Mostly cloudy" and the date/time "1:23 AM 4/13/2023".

The image displays two screenshots of the Databricks workspace interface, showing the process of creating and persisting a table named 'Orders_csv'.

Top Screenshot: The 'Orders' table is displayed with the following data:

	order_id	customer_id	order_date	total_order_amount
1	1001	1	3/1/2022	40
2	1002	2	3/2/2022	40
3	1003	3	3/2/2022	160

Below the table, the command history shows the execution of a Python command to create a temporary view:

```
1 # Create a view or table
2
3 temp_table_name = "Orders_csv"
4
5 df.createOrReplaceTempView(temp_table_name)
```

The command took 0.14 seconds to execute.

Bottom Screenshot: The 'Orders' table is displayed again, showing the same data. The command history shows the execution of a SQL command to query the temporary view:

```
1 %sql
2
3 /* Query the created temp table in a SQL cell */
4
5 select * from 'Orders_csv'
```

The command took 0.35 seconds to execute. Below the table, the command history shows the execution of a Python command to create a permanent table:

```
1 # With this registered as a temp view, it will only be available to this particular notebook. If you'd like other users to be able to query this
2 # table, you can also create a table from the DataFrame.
3 # Once saved, this table will persist across cluster restarts as well as allow various users across different notebooks to query this data.
4 # To do so, choose your table name and uncomment the bottom line.
5 permanent_table_name = "Orders_csv"
6
7 # df.write.format("parquet").saveAsTable(permanent_table_name)
```

The command took 0.09 seconds to execute.

Here I uncommented the line 7 in command 5 shells for all the notebooks to make sure the tables I have created in databricks dbfs can be accessed in any notebook

- For customers table

community.cloud.databricks.com/?o=7575967165101867#notebook/2666332837658399/command/2666332837658404

Customers Python

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```
5 select * from 'Customers_csv'
```

(1) Spark Jobs

_sqldf: pyspark.sql.dataframe.DataFrame = [customer_id: string, name: string ... 3 more fields]

	customer_id	name	email	phone	location
1	1	John Doe	johndoe@example.com	555-555-5555	New York
2	2	Jane Doe	janedoe@example.com	555-555-5556	Los Angeles
3	3	Bob Smith	bobsmith@example.com	555-555-5557	Chicago

3 rows | 0.52 seconds runtime Refreshed 28 minutes ago

Command took 0.52 seconds -- by venkatagayathrip149@gmail.com at 4/13/2023, 1:09:59 AM on gayathri1

```
Cmd 5
1 # With this registered as a temp view, it will only be available to this particular notebook. If you'd like other users to be able to query this
2 # table, you can also create a table from the DataFrame.
3 # Once saved, this table will persist across cluster restarts as well as allow various users across different notebooks to query this data.
4 # To do so, choose your table name and uncomment the bottom line.
5 permanent_table_name = "Customers_csv"
6 df.write.format("parquet").saveAsTable(permanent_table_name)
```

(1) Spark Jobs

Command took 7.66 seconds -- by venkatagayathrip149@gmail.com at 4/13/2023, 1:14:24 AM on gayathri1

Shift+Enter to run
Shift+Ctrl+Enter to run selected text

community.cloud.databricks.com/?o=7575967165101867#notebook/2666332837658418/command/2666332837658419

MyLabApril15 SQL

File Edit View Run Help Last edit was 1 minute ago Give feedback

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```
Cmd 1
1 /*basic syntax to check if i'm able to access the table in another notebook*/
2 SELECT COUNT(*) FROM Customers_csv
3
```

(2) Spark Jobs

count(1)
3

1 row | 0.95 seconds runtime Refreshed now

Command took 0.95 seconds -- by venkatagayathrip149@gmail.com at 4/13/2023, 1:39:37 AM on gayathri1

Shift+Enter to run
Shift+Ctrl+Enter to run selected text

Task 2

community.cloud.databricks.com/?o=7575967165101867#notebook/2666332837658418/command/2666332837658421

MyLabApril15 SQL

File Edit View Run Help Last edit was 2 minutes ago Give feedback

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Cmd 1

```
1 /*basic syntax to check if i'm able to access the table in another notebook*/
2 /*SELECT COUNT(*) FROM Customers_csv*/
3 /*task 1 imported all datasets into databricks*/
4 /*task2 : Perform an aggregation to determine the total number of orders and total revenue
5 generated by the online retail store.*/
6 SELECT COUNT(DISTINCT order_id) AS total_orders, SUM(quantity * price) AS total_revenue
7 FROM OrderItems1_csv;
```

3 Spark Jobs

Table +

	total_orders	total_revenue
1	3	240

1 row | 1.47 seconds runtime Refreshed 1 minute ago

Command took 1.47 seconds -- by venkatagayathrip1498@gmail.com at 4/13/2023, 9:30:11 AM on My Cluster

Task 3

Cmd 2

```
1 /*task 3 : Join the Orders and Customers tables to create a new table that contains the name and
2 email address of each customer who placed an order*/
3 CREATE TABLE Cust_with_Orderz AS
4 SELECT DISTINCT c.name, c.email
5 FROM Orders1_csv o
6 JOIN Customers1_csv c ON o.customer_id = c.customer_id
7 JOIN OrderItems1_csv oi ON o.order_id = oi.order_id
8 WHERE oi.quantity > 0;
9
```

8 Spark Jobs

Query returned no results

Command took 7.39 seconds -- by venkatagayathrip1498@gmail.com at 4/13/2023, 9:30:32 AM on My Cluster

Cmd 3

```
1 SELECT * FROM Cust_with_Orderz
```

2 Spark Jobs

Table +

	name	email
1	Jane Doe	janedoe@example.com
2	John Doe	johndoe@example.com
3	Bob Smith	bobsmith@example.com

3 rows | 1.52 seconds runtime Refreshed now

Command took 1.52 seconds -- by venkatagayathrip1498@gmail.com at 4/13/2023, 9:30:38 AM on My Cluster

Task 4

MyLabApril15 SQL

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Run all My Cluster Publish

Cmd 4

```
1 /*task 4 : Join the Order Items and Products tables to create a new table that contains the name and
2 price of each product included in an order.*/
3 CREATE TABLE Product_Prices AS
4 SELECT DISTINCT p.product_name, p.price,o.order_id
5 FROM Products p
6 JOIN OrderItems o ON p.product_id=o.product_id
```

▶ (7) Spark Jobs

Query returned no results

Command took 29.62 seconds -- by venkatagayathrip1498@gmail.com at 4/13/2023, 11:51:48 PM on My Cluster

Cmd 5

```
1 SELECT * FROM Product_Prices order by price asc;
```

▶ (1) Spark Jobs

Table +

	product_name	price	order_id
1	T-shirt	20	1001
2	T-shirt	20	1003
3	Jeans	40	1002
4	Sneakers	50	1003

4 rows | 1.33 seconds runtime Refreshed now

Command took 1.33 seconds -- by venkatagayathrip1498@gmail.com at 4/13/2023, 11:52:32 PM on My Cluster

Task 5

community.cloud.databricks.com/?o=7575967165101867#notebook/2666332837658418/command/881374398360173

MyLabApril15 SQL

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Run all My Cluster Publish

4 rows | 2.23 seconds runtime Refreshed 18 minutes ago

Command took 2.23 seconds -- by venkatagayathrip1498@gmail.com at 4/13/2023, 8:47:04 AM on My Cluster

Cmd 6

```
1 /*task 5 : Perform an aggregation to determine the total number of products sold by category*/
2 SELECT p.category, COUNT(o.order_id) AS total_products_sold
3 FROM OrderItemsI_csv o
4 JOIN ProductsI_csv p ON o.product_id = p.product_id
5 GROUP BY p.category;
```

▶ (3) Spark Jobs

Table +

	category	total_products_sold
1	Clothing	3
2	Shoes	1

2 rows | 2.25 seconds runtime Refreshed 1 minute ago

Command took 2.25 seconds -- by venkatagayathrip1498@gmail.com at 4/13/2023, 9:03:59 AM on My Cluster

Shift+Enter to run
Shift+Ctrl+Enter to run selected text

Task 6

community.cloud.databricks.com/?o=7575967165101867#notebook/2666332837658418/command/881374398360174

MyLabApril15 SQL

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Run all My Cluster

Command took 2.25 seconds -- by venkatagayathrip1498@gmail.com at 4/13/2023, 9:03:59 AM on My Cluster

Cmd 7

```
1 /*task 6 :Perform a join to create a new table that contains the name, email address, and total
2 revenue generated by each customer who placed an order.*/
3 CREATE TABLE Cust_Revenue AS
4 SELECT DISTINCT c.name, c.email,o.total_order_amount as total_revenue
5 FROM Orders1_csv o
6 JOIN Customers1_csv c ON o.customer_id = c.customer_id
7
8
```

(7) Spark Jobs

Query returned no results

Command took 7.88 seconds -- by venkatagayathrip1498@gmail.com at 4/13/2023, 9:17:36 AM on My Cluster

Cmd 8

```
1 SELECT * FROM Cust_Revenue
```

(2) Spark Jobs

Table +

	name	email	total_revenue
1	Jane Doe	janedoe@example.com	40
2	Bob Smith	bobsmith@example.com	160
3	John Doe	johndoe@example.com	40

3 rows | 1.34 seconds runtime Refreshed 1 minute ago

Command took 1.34 seconds -- by venkatagayathrip1498@gmail.com at 4/13/2023, 9:17:48 AM on My Cluster

Shift+Enter to run
Shift+Ctrl+Enter to run selected text

Task 7

community.cloud.databricks.com/?o=7575967165101867#notebook/2666332837658418/command/881374398360176

MyLabApril15 SQL

File Edit View Run Help Last edit was 3 minutes ago Give feedback

Run all My Cluster Publish

Cmd 9

```
1 /*task 7 Perform a join to create a new table that contains the name and category of each product
2 included in an order.*/
3 CREATE TABLE order_products AS
4 SELECT o.order_id, p.product_name, p.category
5 FROM Orders1_csv o
6 JOIN OrderItems1_csv oi ON o.order_id = oi.order_id
7 JOIN Products1_csv p ON oi.product_id = p.product_id;
8
```

(7) Spark Jobs

Query returned no results

Command took 6.80 seconds -- by venkatagayathrip1498@gmail.com at 4/13/2023, 9:23:38 AM on My Cluster

Cmd 10

```
1 SELECT * FROM order_products
```

(2) Spark Jobs

Table +

	order_id	product_name	category
1	1001	T-shirt	Clothing
2	1002	Jeans	Clothing
3	1003	T-shirt	Clothing
4	1003	Sneakers	Shoes

4 rows | 1.45 seconds runtime Refreshed 2 minutes ago

Command took 1.45 seconds -- by venkatagayathrip1498@gmail.com at 4/13/2023, 9:23:58 AM on My Cluster

Shift+Enter to run
Shift+Ctrl+Enter to run selected text

Task 8

Cmd 11

```
1  /*task Perform an aggregation to determine the total revenue generated by each product
2  category.*/
3  SELECT p.category, SUM(oi.quantity * p.price) AS total_revenue
4  FROM orderItems1_csv oi
5  JOIN Products1_csv p ON oi.product_id = p.product_id
6  GROUP BY p.category;
```

- ▶ (3) Spark Jobs

Table

	category	total_revenue
1	Clothing	140
2	Shoes	100

2 rows | 1.95 seconds runtime

Refreshed now

Command took 1.95 seconds -- by venkatagayathrip1498@gmail.com at 4/13/2023, 9:29:29 AM on My Cluster

Shift+Enter to run

Shift+Ctrl+Enter to run selected text